

# AKSHAY CHOUDHARY

☎ (+91) 8827406511 ✉ [akshaychoudhary2040@gmail.com](mailto:akshaychoudhary2040@gmail.com) [in linkedin](#) [github](#) [codechef](#) [leetcode](#)

## EDUCATION

National Institute of Technology, Rourkela

Bachelor of Technology in Biotechnology | CGPA: 8.58/10

Aug 2023 – Present

Rourkela, Odisha

## TECHNICAL SKILLS

**Languages:** Python, C, C++, SQL **AI/ML:** Scikit-Learn, Hugging Face Transformers, LangChain, NLP, LLM

Fine-tuning, RAG Pipelines

**Backend & Development:** FastAPI, React.js, Docker, REST APIs, Streamlit, Git/GitHub

**Core CS:** Data Structures and Algorithms, Operating Systems, Computer Networks, DBMS

**Systems:** Linux, POSIX Threads, Socket Programming, Multithreading

**Developer Tools:** Git/GitHub, VS Code, PyCharm, Jupyter Notebook

## PROJECTS

**AI Research Paper Summarizer** | *LangChain, Streamlit, Groq LLM*

[Github](#)

- Built an LLM-powered research paper summarization platform using LangChain and Groq APIs to generate structured summaries with configurable tone, depth, and output formats.
- Engineered document ingestion and prompt orchestration pipelines for extracting key insights from technical research papers.
- Developed an interactive Streamlit interface enabling faster literature review workflows for long-form academic content.
- Optimized prompt design and response generation workflows to improve summary consistency and readability across research domains.

**AI-Powered Online Meeting Summarizer** | *Pegasus LLM, Hugging Face, React, FastAPI*

[Github](#)

- Developed an end-to-end AI meeting summarization system using Pegasus transformers, FastAPI, and React for automated meeting transcription and abstractive summarization.
- Built REST APIs for audio upload, preprocessing, and real-time summary generation workflows.
- Integrated Hugging Face transformer pipelines for generating concise meeting notes and action-item summaries from long-form discussions.
- Designed scalable frontend-backend interaction workflows supporting asynchronous processing and user-driven summary retrieval.

**Multithreaded HTTP Server in C** | *C, POSIX Sockets, Multithreading, Linux*

[Github](#)

- Built a multithreaded HTTP server in C using POSIX sockets and pthreads capable of handling concurrent client connections on Linux systems.
- Implemented thread-pool based request scheduling using mutexes, condition variables, and producer-consumer synchronization mechanisms.
- Designed low-level HTTP request parsing and concurrent response handling pipelines following HTTP protocol standards.
- Strengthened understanding of systems programming concepts including concurrency, synchronization, socket programming, and resource management.

## OPEN SOURCE CONTRIBUTIONS

**PyCQA/Pylint Contributor** | *Python, Static Analysis, Open Source*

[Github](#)

- Contributed production-grade fixes to improving Python standard-library compatibility and static analysis reliability.
- Resolved function redefinition edge cases and debugging issues through merged in PR 10595 and PR 10606.
- Collaborated with maintainers through code reviews, issue discussions, and iterative patch refinement in a large-scale open-source codebase.

## ACHIEVEMENTS

- **Top 10 Finalist – Teach-a-thon Hackathon** among 500+ teams for solving real-world challenges under time constraints.
- **Solved 250+ DSA problems** across LeetCode and CodeChef platforms.
- **LeetCode Max Rating: 1497** demonstrating strong problem-solving and analytical skills.
- **CodeChef 2-Star (Max Rating: 1483)** with consistent participation in competitive programming contests.

## LEADERSHIP

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### Hackathon Organizer

*HackOdisha 4.0 – National Institute of Technology, Rourkela*

**August 2024**

*Rourkela, Odisha*

- Coordinated a 36-hour national-level hackathon with 100+ participating teams and Collaborated with organizing teams to manage technical operations, participant coordination, and event execution for 100+ teams.